MIL-B-49430/1E(ER)
25 April 1988
SUPERSEDING
MIL-B-49430/1D(ER)
15 June 1987

MILITARY SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE BA-5599/U

This specification sheet is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the lithium sulfur dioxide non-rechargeable battery described herein shall consist of this document and the latest issue of MIL-B-49430(ER).

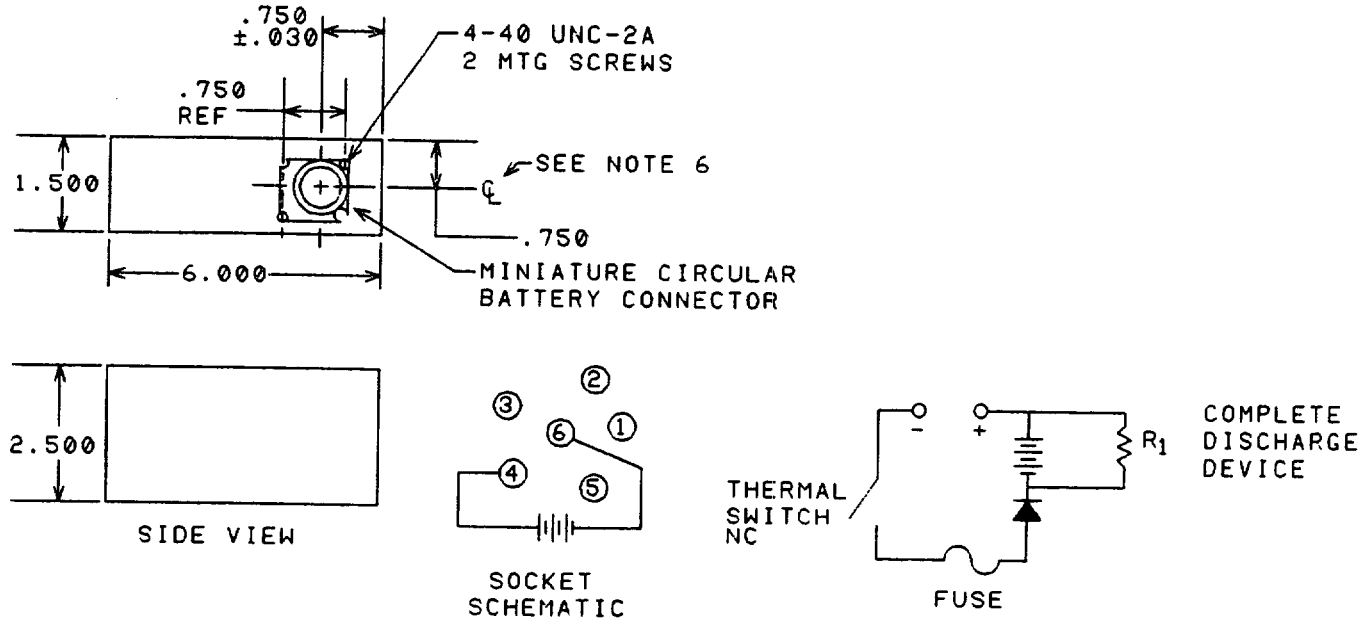


Figure 1. Battery dimensions and wiring schematic.

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NOTES:

1. All dimensions are in inches.

2. Tolerences, unless otherwise specified, are -.063,+.000.

3. Socket wiring schematic to be printed or embossed on jackets in suitable location near the socket on the terminal face of the battery.

4. The battery terminal shall consist of a miniature circular battery connector, Dwg SC-C-179492 or equal. Connector shall be mounted in accordance with referenced drawing.

5. Connector socket body configuration shall accept two (2) each 0.156 inch diameter guideposts as shown. Depth of cavity shall be 0.622 inch.

6. Connector shall be located on centerline.

7. The socket shall be supported and mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the jacket and shall not be more than 1/16 inch below adjacent outside surface of the jacket when used initially, during, and after subsequent insertions of the mating plug during contractual testing.

8. R_1 value shall be between 90 ohms and 125 ohms and have a power rating of 1.0 watt.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 9.15 volts.

Minimum voltage for capacity: 6.00 volts

The battery shall contain a 2.25 ampere non-replaceable slow blow fuse in the negative leg.

The battery shall contain a diode wired in series. See paragraph 3.5.5.2 of the basic specification.

Terminals: Six (6) hole socket type with no obstructions of any holes (see note 5).

Weight (maximum): 1.0 pound (454 grams)

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Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity Requirements in Hours to 6.00 Volts
I	3.0
L	1.6
H	5.6
HT	5.4
LT	1.4
T	2.8
D	2.8
ILR	1000

Initial voltage delay: During the I, H, HT, T, D and ILR discharge tests, initial voltages below 6.0 volts cannot exceed a 5.0 second duration. During the L and LT discharge tests inital voltages below 5.70 volts cannot exceed a 60.0 second duration.

High temperature switch: A high temperature switch or thermal fuse shall be employed internal to the battery as described in Para. 3.5.5 and 3.5.5.1 of the basic specification.

Complete discharge device: This device shall be inserted in the battery as shown on the circuit schematic and shall have the characteristics as described in para 3.5.5.3 of the basic specification.

Connector: The connector shall be in accordance with SC-C-179492 with the following exception: The durability test described in paragraph 4h. Shall be performed 75 times with connector SC-D-882165, 75 times with connector SC-C-179493, and 100 times with SM-D-687888. Compliance with this requirement shall be demonstrated as a part of incoming inspection.

METHODS OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage: Details on storage conditions for all capacity tests are specified in the basic specification (see 4.8.9).

(2) Discharge:

Cells: Discharge at 2.0 amp constant current at $70 \pm 5^{\circ}$ for a minimum of 2.7 hours to 2.00 volts. Continuous discharge to zero volts. Then discharge at 1.0 ampere constant current for 5.4 hours into voltage reversal.

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Batteries: For all capacity tests except the H, HT, and ILR, the batteries shall be discharged at two amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged through two amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall repeated until the closed circuit voltage reaches zero volts. There shall be no venting, fire or explosion when the batteries are force discharged. For the ILR test a fixed resistance of 1575 ohms shall be used to cut-off voltage of 6.00 volts or 1000 hours.

Closed circuit voltage test: Load resistance of 8 ohms shall be used. Voltage shall be above 6.00 volts with 5.0 seconds.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodian:
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(Project 6135-A209-01)

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